

Lecture 6: Regularization, Ridge and Lasso Regression

ML/AI for Engineering Applications
ENV 6932

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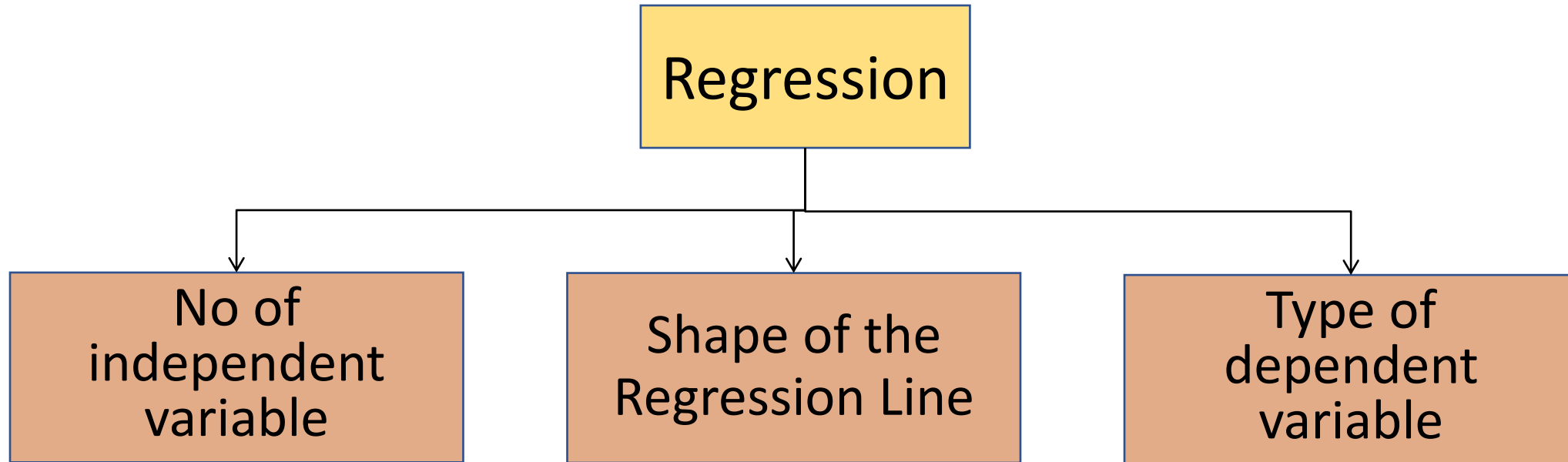
Outline

- Bias-Variance Tradeoff
- Regularization
- Ridge Regression
- Lasso Regression

Types of Regressions

- Linear Regression
- Logistic Regression
- Polynomial Regression
- Stepwise Regression
- Ridge Regression
- Lasso Regression
- ElasticNet Regression

How to determine the proper regression type?

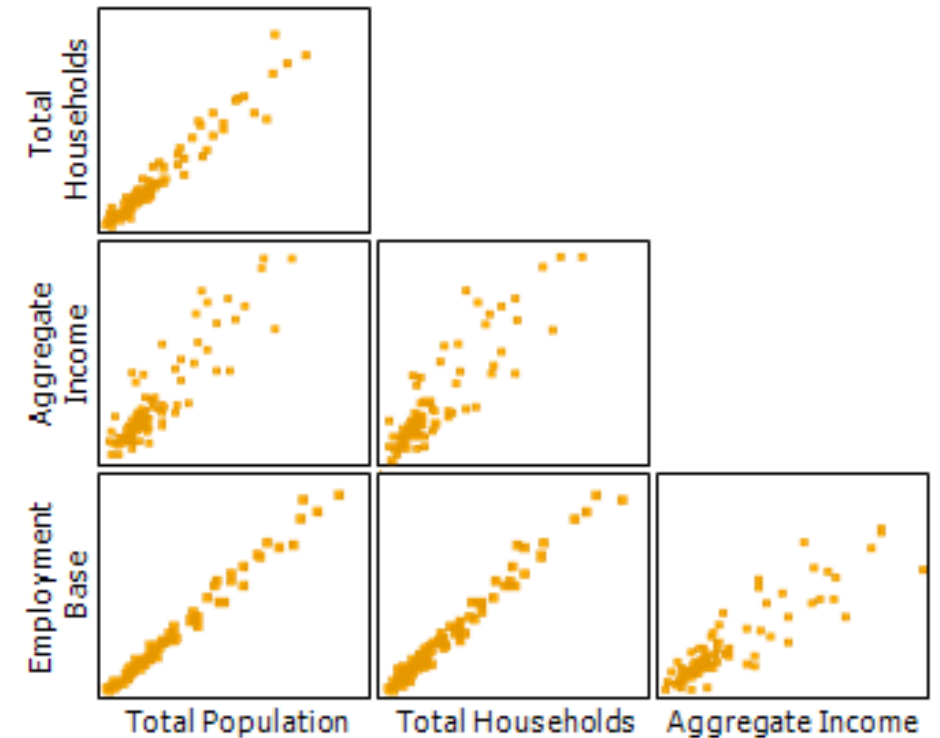


Important Points: Linear Regression

- There must be **linear relationship** between predictors and dependent variables
- Multiple regression suffers from **multicollinearity, autocorrelation, heteroskedasticity**.

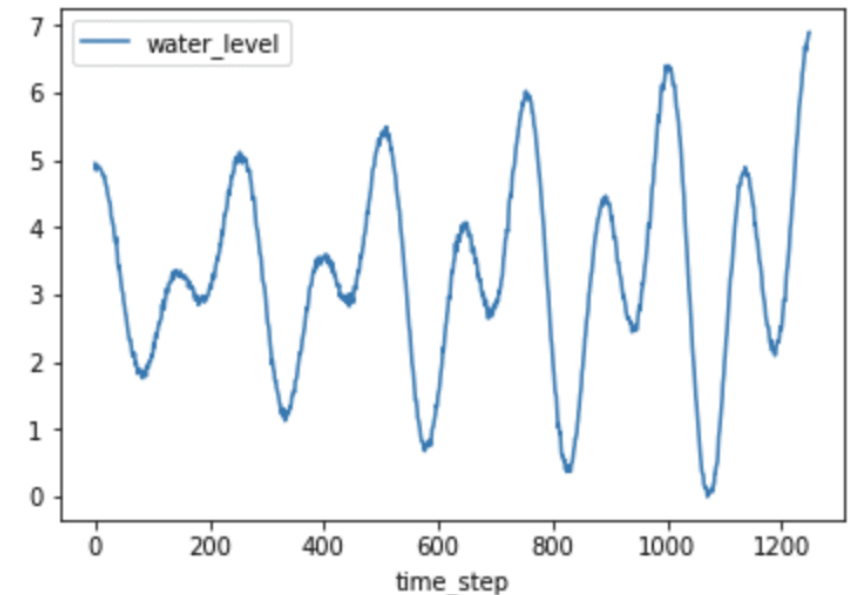
Important Points: Linear Regression

- **Multicollinearity:** one predictor variable can be linearly predicted from the others with a substantial degree of accuracy.
- **Solution:**
- Create a correlation matrix in order to check for multicollinearity.
- Drop the highly correlated variables
- Use principal component analysis, which can help to reduce the number of variables and increase independent terms.



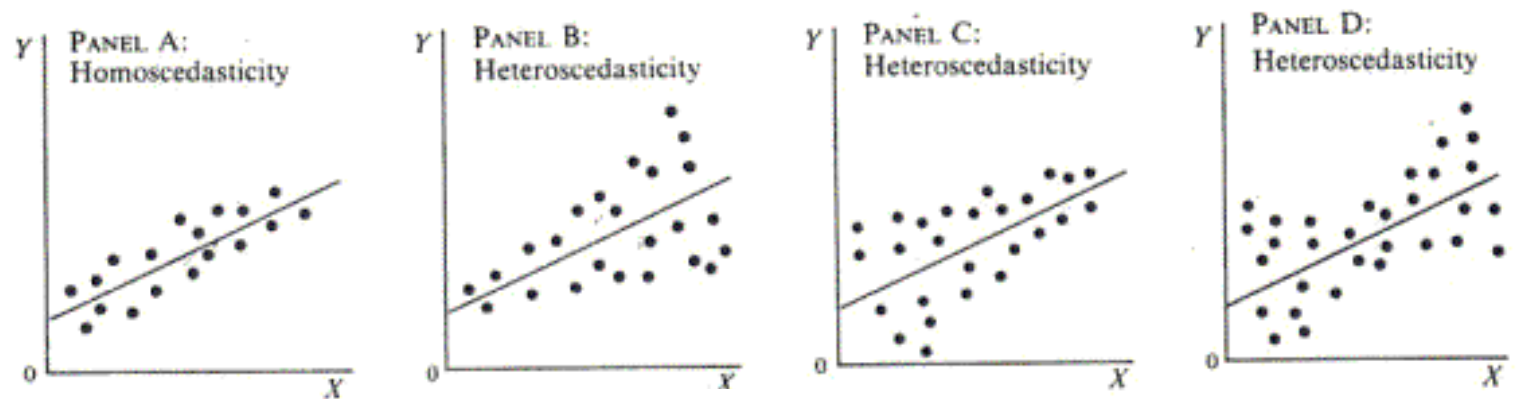
Important Points: Linear Regression

- **Autocorrelation:** or serial correlation, is the correlation of a signal with a delayed copy of itself as a function of delay. There is nonzero correlation between variables at different time point.
- **Solution:** remove seasonality



Important Points: Linear Regression

- **Heteroskedasticity:** means not all of the random variables have the same variance, situations where the variance of the residuals is unequal over a range of measured values
- **Solution:** Re-build the model with new predictors.
Variable transformation (e.g. Box-Cox transformation)



Important Points: Linear Regression

- **Multicollinearity:** one predictor variable can be linearly predicted from the others with a substantial degree of accuracy.
- **Autocorrelation:** or serial correlation, is the correlation of a signal with a delayed copy of itself as a function of delay. There is nonzero correlation between variables at different time point.
- **Heteroskedasticity:** means not all of the random variables have the same variance, situations where the variance of the residuals is unequal over a range of measured values

Important Points: Linear Regression

- There must be **linear relationship** between independent and dependent variables
- Multiple regression suffers from **multicollinearity, autocorrelation, heteroskedasticity**.
- Linear Regression is very sensitive to **Outliers**.
- Multicollinearity can increase the variance of the coefficient estimates and make the estimates very sensitive to minor changes in the model.
- In case of multiple independent variables, most significant independent variables can be selected with **forward selection, backward elimination** and **step wise approach**.

Important Points: Logistic Regression

- Widely used for **classification problems**
- Logistic regression doesn't require linear relationship, it can handle various types of relationships as it uses a non-linear log transformation.
- To avoid over fitting and under fitting, include all significant variables. A good approach is to use a step wise method.

Important Points: Logistic Regression

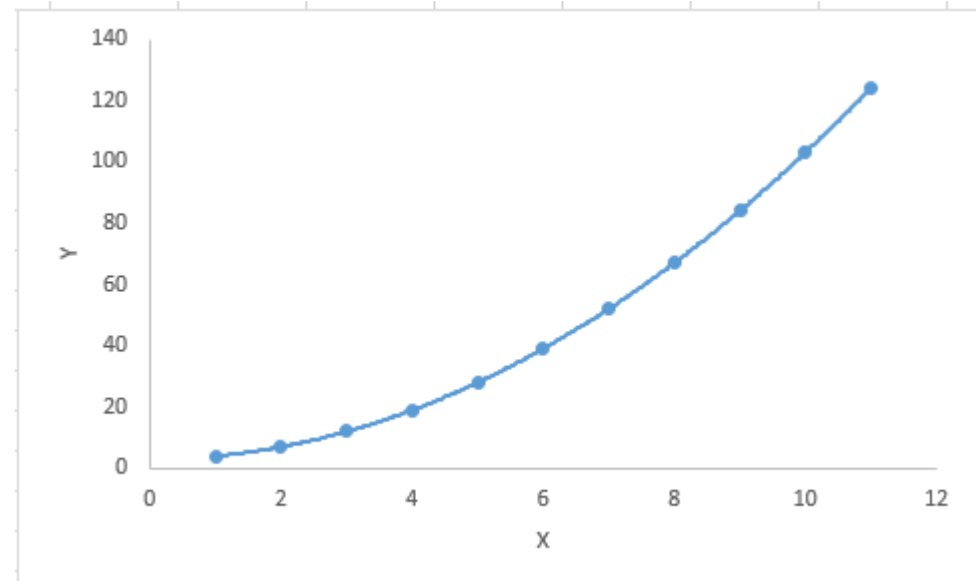
- It needs **large sample sizes** as maximum likelihood estimates are less powerful at low sample sizes than ordinary least square.
- The independent variables should not be correlated with each other i.e. **no multicollinearity**. However, interaction effects of categorical variables can be included in the analysis and in the model.

Important Points: Logistic Regression

- **Ordinal logistic regression:** If the values of dependent variable is ordinal.
- **Multinomial Logistic regression:** If dependent variable is multi class then it is known as Multinomial Logistic Regression.

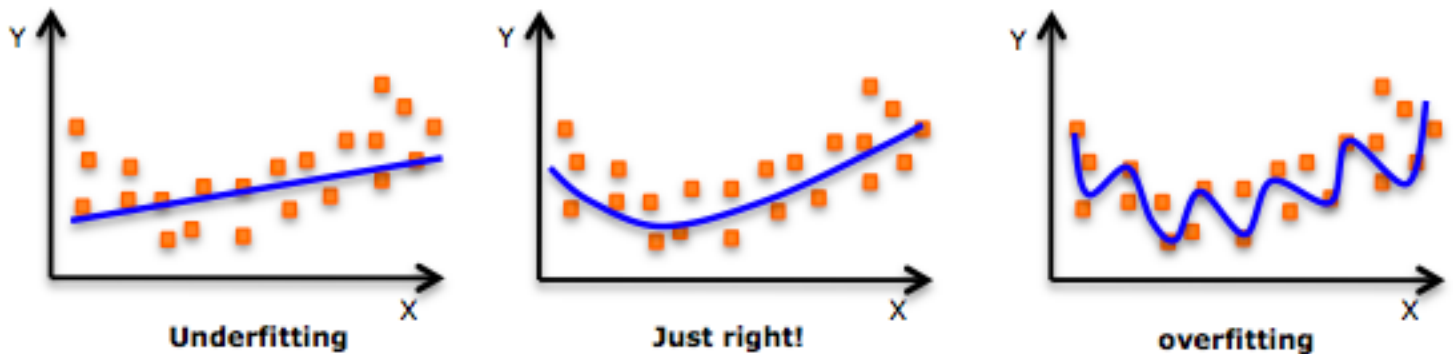
Polynomial Regression

- Polynomial regression: if the power of independent variable is more than 1.



Important Points: Polynomial Regression

- While there is an inclination to fit a higher degree polynomial to get lower error, this can result in over-fitting.
- Always plot to see the fit and make sure that the curve fits the nature of the problem.
- Especially look out for curve towards the ends and see whether those shapes and trends make sense.



Bias-Variance Tradeoff

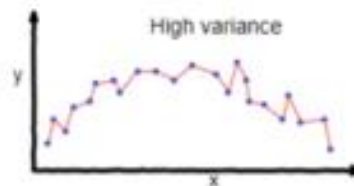
- A big portion of constructing the best models in ML deals with the bias-variance tradeoff.
- **What is Bias?** refers to how correct (or incorrect) the model is.
- A very simple model that makes a lot of errors has high bias.
- A very complicated model that performs well on its training data has low bias.

Bias-Variance Tradeoff

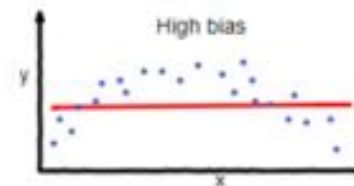
- **What is Variance?**
- Variance of a model is negatively correlated with bias
- It shows how much a prediction could potentially vary if one of the predictors changes slightly.
- In a **simple model**, the simplicity of the model makes its predictions change slowly with predictor value, results into **low variance**.
- A complicated, low bias model likely performs very well with the training data and so predictions vary largely as predictor values change a little. **A complicated model has high variance**, and it will not generalize to new data well.

Bias-Variance Tradeoff

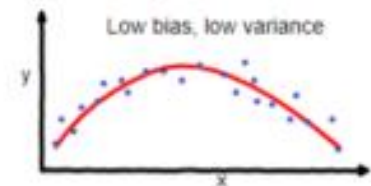
- The low-bias/high-variance model displays overfitting.
- In this case, the model has too many terms and explains random noise in the data on top of the overall trend. This results into low performance on unseen/new data.
- The high-bias/low-variance model shows underfitting.
- In this case, the model is too simple (too few terms) to suitably describe the trend seen in the data. Again, the model does not perform well with new data.



overfitting



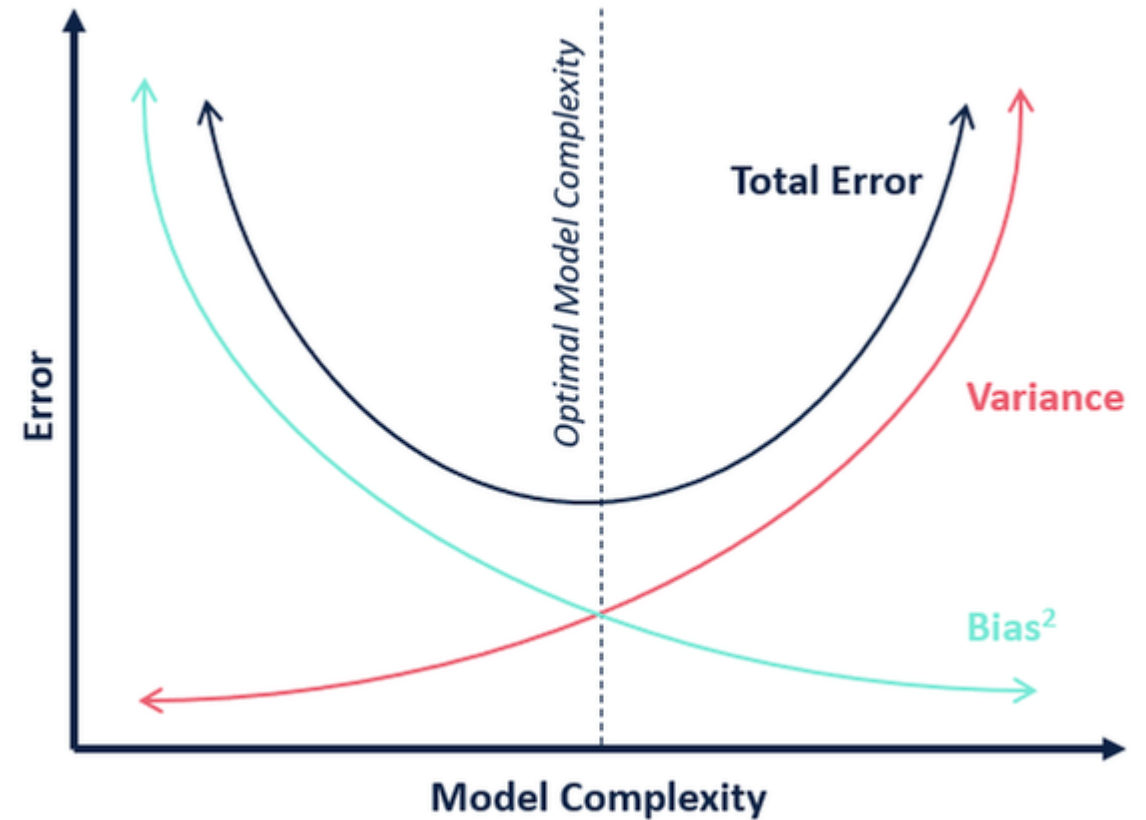
underfitting



Good balance

Bias-Variance Tradeoff

- Therefore, we need to have **the appropriate number of terms** to define the trend without fitting to the noise.
- We need some sort of **feature selection** in which predictors with no relationship with the dependent variable are not dominant in the final model.
- The total error consist of three terms: the $(\text{bias})^2$, the variance, and an irreducible error term.



Model Performance

- How well is performing a model on the new dataset?
- Let's calculate **the predicted error**: the difference between the actual y and the predicted y from the new dataset.

- The expected squared predicted error at a point x :

$$\text{Err}(x) = E [(Y - \hat{f}(x))^2]$$

This error can be decomposed into bias and variance as:

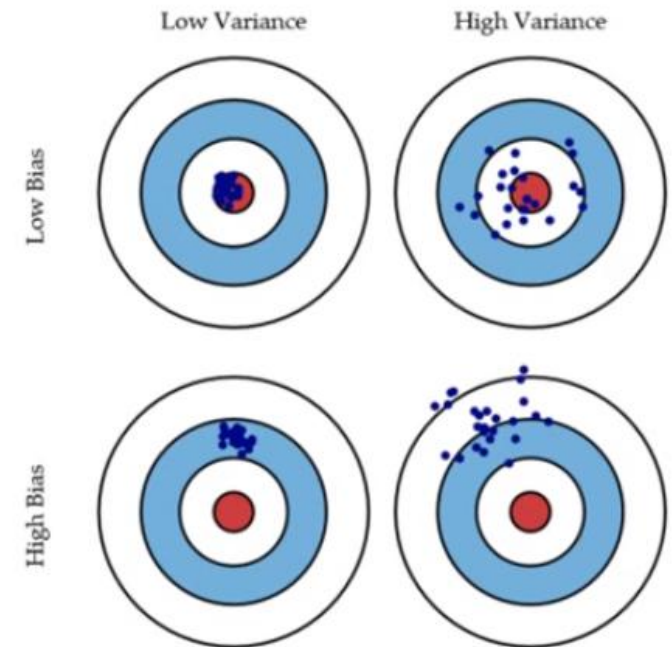
$$\text{Err}(x): (E[\hat{f}(x)] - f(x))^2 + E[(\hat{f}(x) - E[\hat{f}(x)])^2] + \sigma^2$$

Where $f(x) = Y$

Predicted Error: $\text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$

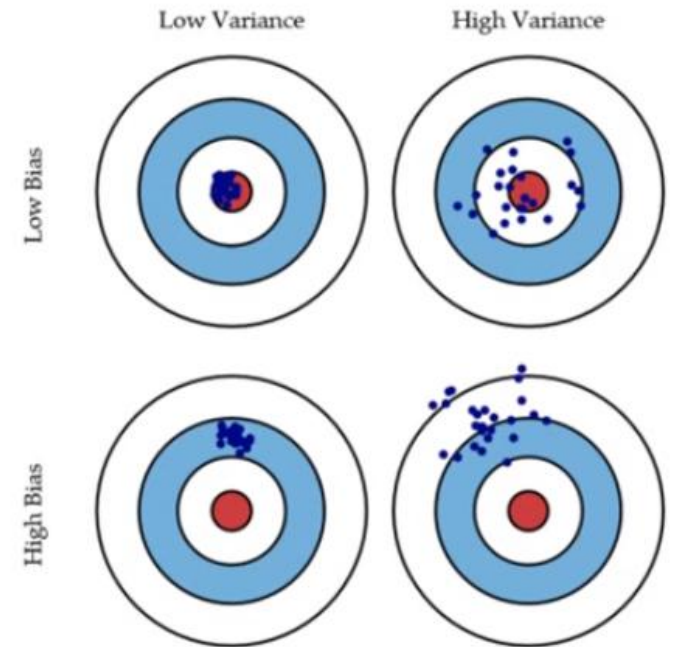
Model Performance

- **The irreducible error:** the inherent uncertainty around the mean.
- We cannot do anything about the inherent uncertainty.
- We need to handle bias and variance of the model.
- **Bias:** simply means how far away is our estimated values from actual values. Bias is an error due to erroneous assumptions in the learning algorithm.



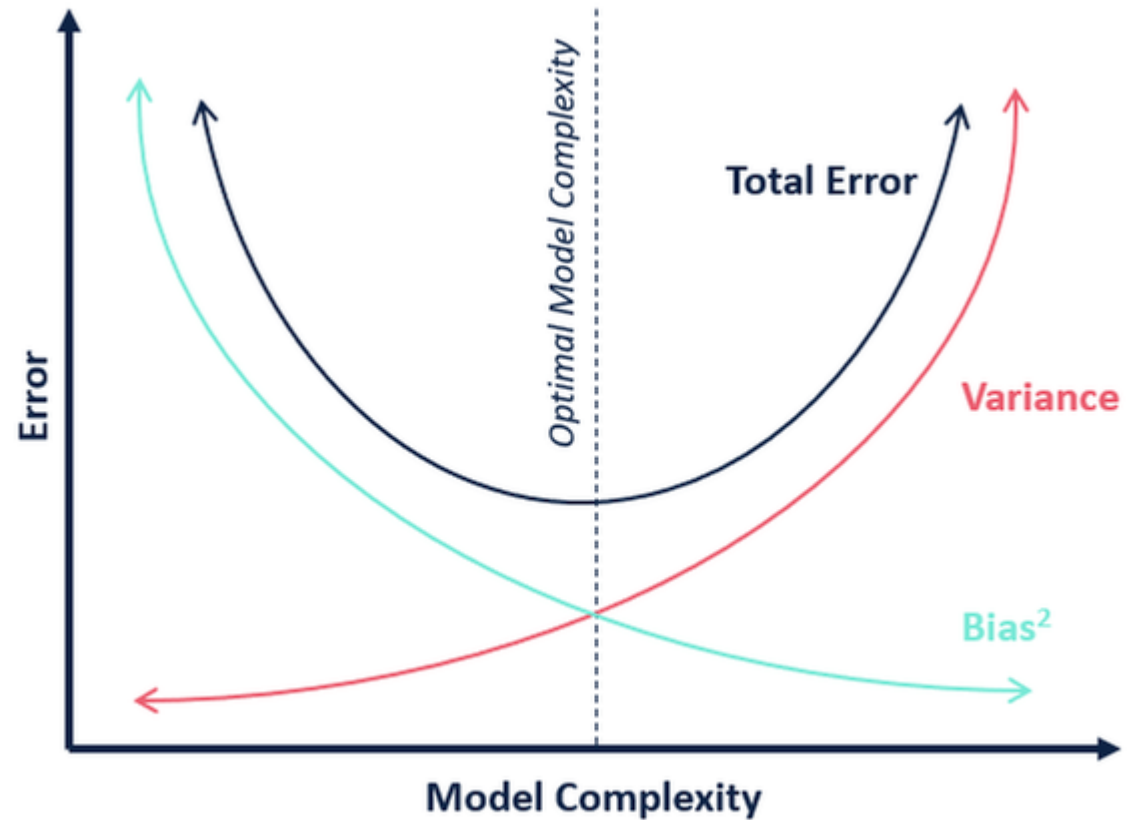
Model Performance

- **Bias:** simply means how far away is our estimated values from actual values. Bias is an error due to erroneous assumptions in the learning algorithm.
- **Variance:** a measure of spread or variations in our predictions, the variance is an error from sensitivity to small variations in the training set.
- High variance can cause an algorithm to model the random noise in the training data (overfitting), rather than the intended outputs.



Bias-Variance Tradeoff

- How to avoid overfitting?
- How to trade some of that variance with bias to reduce our overall error?
- One of the most common methods to avoid overfitting is by reducing the model complexity using regularization.



Ridge Regression

- Is used when the data suffers from multicollinearity (independent variables are highly correlated).
- In multicollinearity, while the least squares estimates are unbiased, their variances are large which differs the observed value from the true value.
- ridge regression reduces the standard errors by adding a degree of bias to the regression estimates.

Ridge Regression

- Ridge regression solves the multicollinearity problem through shrinkage parameter λ

$$= \operatorname{argmin}_{\beta \in \mathbb{R}^p} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{Penalty}}$$

- Ridge regression shrinks the value of coefficients but doesn't reaches zero.
- It uses a regularization method, **l2 regularization**.
- **Regularization:** The process of adding a norm to our cost function.

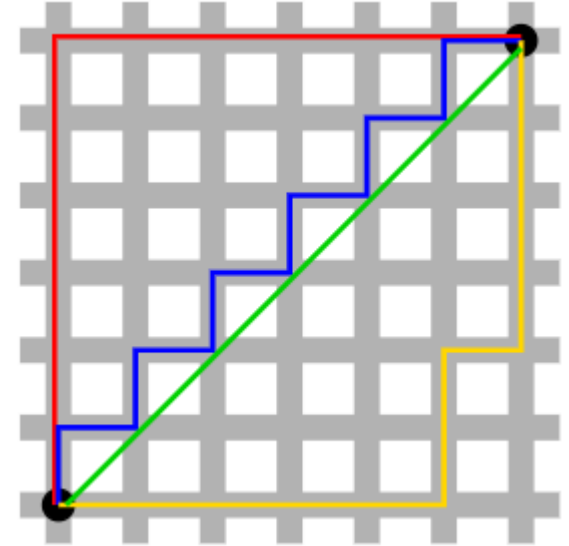
Definition of Norm

- Definition of a p-norm

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{1/p}$$

$$\|x\|_2 = \sqrt{\left(\sum_i x_i^2 \right)} = \sqrt{x_1^2 + x_2^2 + \dots + x_i^2}$$

$$\|x\|_1 = \sum_i |x_i| = |x_1| + |x_2| + \dots + |x_i|$$



What is p-norms and regularization?

- Taxicab distance is the 1-norm, also known as the L1 norm.
- The L2 norm is actually the 2-norm, Euclidian distance, squared.

$$\text{Ridge Cost} = (y - X\beta)^T (y - X\beta) + ||\beta||_2^2$$

$$\text{Lasso Cost} = (y - X\beta)^T (y - X\beta) + ||\beta||_1$$

- This process of adding a norm to our cost function is known as **regularization**.
- The data can be regularized for different underlying reasons and with different effects.
- In ridge and lasso regression, regularizers are built to address colinearity and model complexity problems.

Regularization

- When the β coefficients are unconstrained (OLS: Ordinary Least Square) they can tend to **have high value** (in multicollinear cases), which would result in very high variance.
- To control the variance, we **add a constraint** to the β coefficients when estimating them.
- This constraint is the penalty parameter given by λ (lambda).
- This type of regression where we add penalty parameter λ in order to estimate β is called Ridge and Lasso regression.

Ridge Regression

- Ridge regression solves the multicollinearity problem through shrinkage parameter λ

Linear regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon \quad \leftarrow \text{True Model}$$
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p \quad \leftarrow \text{Estimated Model}$$
$$\hat{\beta}^{OLS} = (X^T X)^{-1} X^T Y \quad \leftarrow \text{Matrix Equation}$$

In ridge regression

$$= \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{Penalty}}$$

$$\|B\|_2 = \sqrt{\beta_0^2 + \beta_1^2 + \cdots + \beta_p^2}$$

$$\hat{\beta}^{ridge} = (X^T X + \lambda I)^{-1} X^T Y$$

Review of Linear Regression

$$\sum_{j=1}^p X_{ij}\beta_j = y_i, \quad (i = 1, 2, \dots, n),$$

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{y},$$

$$\mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1p} \\ X_{21} & X_{22} & \cdots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{np} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} S(\boldsymbol{\beta}),$$

$$S(\boldsymbol{\beta}) = \sum_{i=1}^n \left| y_i - \sum_{j=1}^p X_{ij}\beta_j \right|^2 = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

$$(\mathbf{X}^\top \mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}.$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Lasso Regression

- **Lasso:** Least Absolute Shrinkage and Selection Operator.

$$= \operatorname{argmin}_{\beta \in \mathbb{R}^p} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_1}_{\text{Penalty}}$$

- Lasso penalizes the absolute size of the regression coefficients.
- It reduces the variability and improving the accuracy of linear regression models.
- Difference with ridge: Lasso uses absolute values in the penalty function, instead of squares.
- This causes some of the parameter estimates to turn out exactly zero (i.e., variable selection out of given n variables).

Lasso Regression: Important Points

- The assumptions of lasso regression is same as least squared regression except normality is not to be assumed.
- Lasso **helps in feature selection** since it shrinks coefficients to zero (exactly zero)
- Lasso is a regularization method and uses **l1 regularization**
- If group of independent variables are highly correlated, lasso picks **only one of them**.

Ridge Regression

- Ridge regression is an extension of Linear regression.
- In Linear regression, we find the beta coefficients by minimizing the error without any constraints

$$\operatorname{argmin}_{\beta} \sum_{i=1}^N (y_i - (\beta_0 + \sum_{j=1}^p (x_{ij} * \beta_j)))^2$$

- However, in Ridge regression, the Residual Sum Squares is optimized subject to a constraint on the sum of squares of the coefficients:

$$\operatorname{argmin}_{\beta} \sum_{i=1}^N (y_i - (\beta_0 + \sum_{j=1}^p (x_{ij} * \beta_j)))^2$$

subject to the constraint

$$\sum_{j=1}^p \beta_j^2 \leq t$$

Ridge Regression

- The previous optimization model can be written as using Lagrange multiplier:

$$\operatorname{argmin}_{\beta} \left(\lambda \sum_{j=1}^p \beta_j^2 \right) + \sum_{i=1}^N \left(y_i - \left(\beta_0 + \sum_{j=1}^p (x_{ij} * \beta_j) \right) \right)^2$$

- In matrix notation, we have:

$$\hat{\beta}_{ridge} = (X'X + \lambda I_p)^{-1} X'Y$$

Lambda

- λ will always be greater than 0, it controls the amount of shrinkage.
- λ varies between 0 and ∞ but practically between 0 and 1.
- **How to select λ ?** select the value of lambda such that it minimizes the mean squared error.
- Cross-validation: select a grid of λ values and then compute cross-validation error of each λ . Finally, select λ which gives the smallest cross-validation predicted error.
- When $\lambda = 0$, the regression will be similar to linear regression.
- Ridge enforces β to converge to 0 but does not make them 0. it does not enforce the irrelevant variable to become 0 rather, it will reduce their impact.

Lasso Regression

- **Lasso:** Least Absolute Shrinkage and Selection Operator.

$$= \operatorname{argmin}_{\beta \in \mathbb{R}^p} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_1}_{\text{Penalty}}$$

- Lasso penalizes the absolute size of the regression coefficients.
- It reduces the variability and improving the accuracy of linear regression models.
- Difference with ridge: Lasso uses absolute values in the penalty function, instead of squares.
- This causes some of the parameter estimates to turn out exactly zero (i.e., variable selection out of given n variables).

Mathematical equation of Lasso Regression

$$\sum_{i=1}^n (y_i - \sum_j x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

- λ denotes the amount of shrinkage.
- $\lambda = 0$ implies all features are considered and it is equivalent to the linear regression where only the residual sum of squares is considered to build a predictive model
- $\lambda = \infty$ implies no feature is considered i.e, as λ closes to infinity it eliminates more and more features
- The bias increases with increase in λ
- variance increases with decrease in λ

Lasso Original Optimization

$$\begin{aligned}\hat{\beta}^{\text{lasso}} &= \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2, \\ &\text{subject to } \sum_{j=1}^p |\beta_j| \leq t,\end{aligned}$$

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \sum_{n=1}^N (y_n - \beta x_n)^2 + \lambda \sum_{i=1}^p |\beta_i|$$

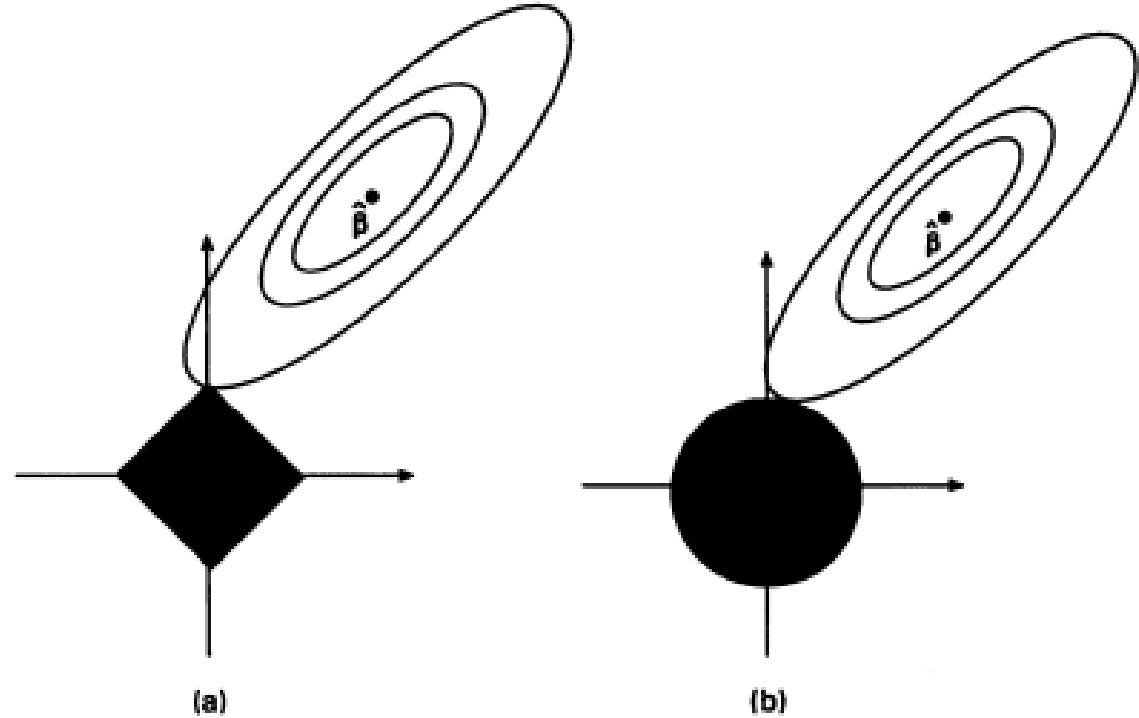
ElasticNet Regression

- hybrid of Lasso and Ridge Regression techniques.
- Uses L1 and L2 prior as regularizer.
- Elastic-net is useful when there are multiple features which are correlated. Lasso is likely to pick one of these at random, while elastic-net is likely to pick both.

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} (\|y - X\beta\|^2 + \lambda_2 \|\beta\|^2 + \lambda_1 \|\beta\|_1).$$

Visual Representation of Lasso and Ridge

- The elliptical contours of the function are shown by the full curves (centered at the OLS estimates);
- The constraint region:
 - **The rotated square for lasso;** The solution is the first place that the contours touch the square, (sometimes occur at a corner, corresponding to a zero coefficient)
 - **circle for ridge regression;** there are no corners for the contours to hit so zero solutions rarely happen



Estimation picture for (a) the lasso and (b) ridge regression

Quick Example: Boston Housing Dataset

- Boston Housing Dataset available in sklearn

Data Set Characteristics:

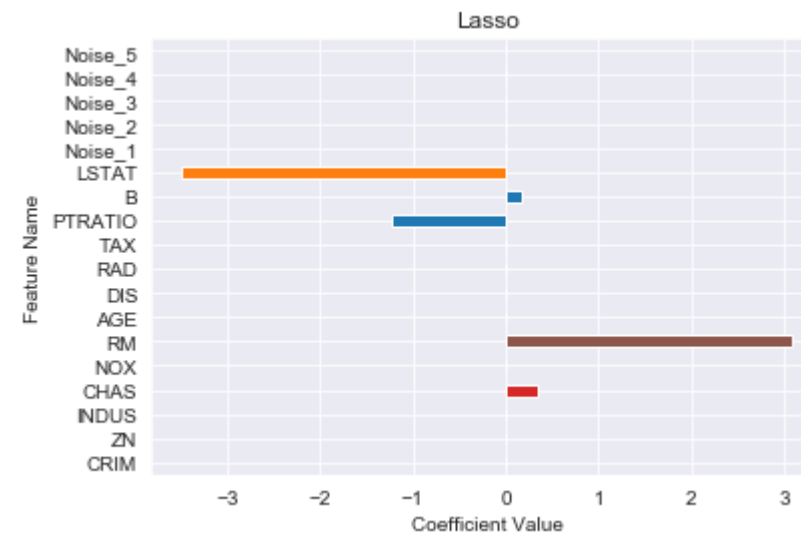
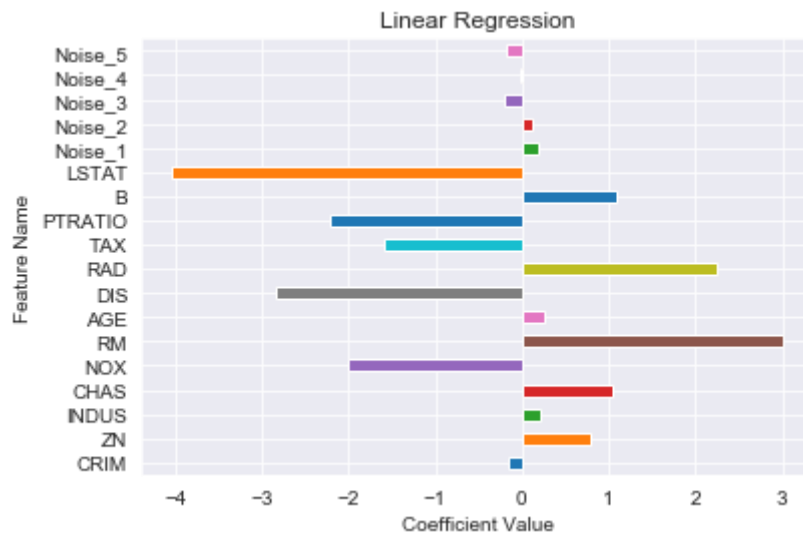
Number of Instances:	506
Number of Attributes:	13 numeric/categorical predictive. Median Value (attribute 14) is usually the target.
Attribute Information (in order):	• CRIM per capita crime rate by town • ZN proportion of residential land zoned for lots over 25,000 sq.ft. • INDUS proportion of non-retail business acres per town • CHAS Charles River dummy variable (= 1 if tract bounds river; 0 otherwise) • NOX nitric oxides concentration (parts per 10 million) • RM average number of rooms per dwelling • AGE proportion of owner-occupied units built prior to 1940 • DIS weighted distances to five Boston employment centres • RAD index of accessibility to radial highways • TAX full-value property-tax rate per \$10,000 • PTRATIO pupil-teacher ratio by town • B $1000(B_k - 0.63)^2$ where B_k is the proportion of black people by town • LSTAT % lower status of the population • MEDV Median value of owner-occupied homes in \$1000's
Missing Attribute Values:	None
Creator:	Harrison, D. and Rubinfeld, D.L.

Quick Example: adding 5 noise features

- Note that in addition to the current features, 5 additional 'features' of random noise was added.
- Let's investigate the ability of all 4 algorithms to filter out irrelevant information.

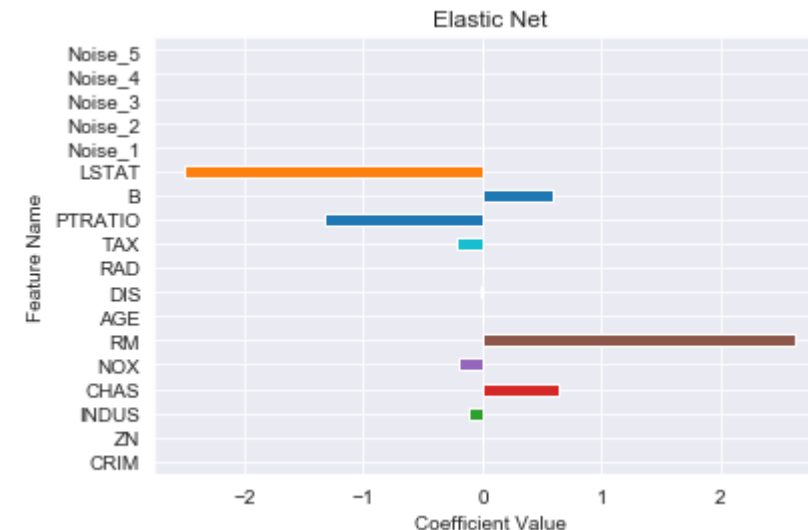
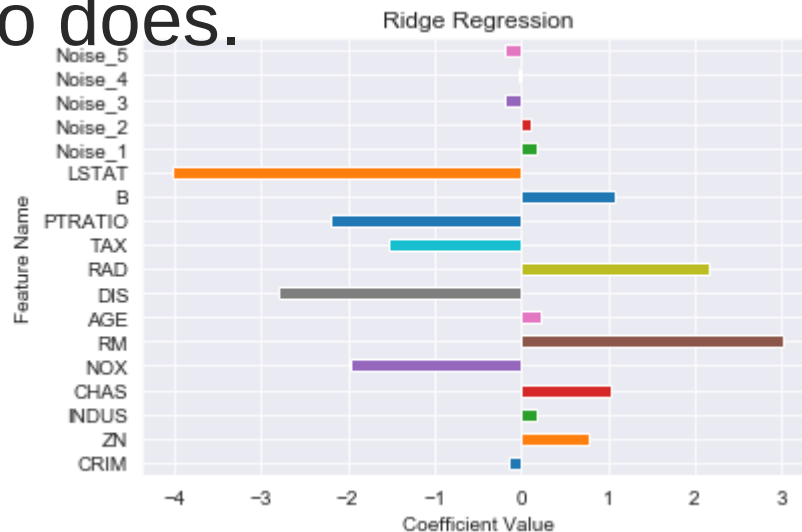
Quick Example: Linear Regression, Lasso

- linear regression assigned non-zero values to all 5 of our noise features, despite none of them having any predictive power.
- Lasso reduces all 5 of our noise features to 0, as well as many of the real features.



Quick Example: Ridge, Elastic Net

- Ridge makes a similar error that unregularized linear regression, assigning coefficient values to our noise features. Also, some features have very small coefficients.
- Similar to Lasso, Elastic Net makes the coefficients of several features 0. However, it does not make as many coefficients 0 as Lasso does.



Quick Example

- For this example, Ridge Regression was the best model based on MSE.
- Although counter-intuitive, it is important to remember the ridge regression **traded some variance for bias**, which ultimately lead to an overall smaller error.
- Lasso and Elastic Net traded **a significant amount of variance for bias**, as a result the error has increased.

```
Mean Squared Error of Linear Regression: 23.87  
Mean Squared Error of Lasso: 24.91  
Mean Squared Error of Ridge Regression: 23.87  
Mean Squared Error of Elastic Net: 24.63
```

Bayesian Regression

- Recall simple linear regression:

$$y_i = \alpha + \beta x_i + \varepsilon$$

- Let's write it in terms of the probabilistic model behind it

$$\mu_i = \alpha + \beta x_i$$

$$y_i \sim \mathcal{N}(\mu_i, \sigma)$$

- i.e. Y follows normal distribution parametrized by μ_i (a linear function of X parametrized by α, β), and by standard deviation σ .
- If we estimate using ordinary least squares, no need to consider probabilistic formulation, since we want to find optimal values of α, β by minimizing the squared errors.
- On another hand, we could estimate such model using MLE (maximizing the likelihood function):

$$\arg \max_{\alpha, \beta, \sigma} \prod_{i=1}^n \mathcal{N}(y_i; \alpha + \beta x_i, \sigma)$$

Density Estimation

- density estimation refers to estimating the probability distribution (or density function) of a random variable based on observed data.

Maximum Likelihood Estimation

- **Maximum likelihood estimation (MLE)** is a method of **estimating** the **parameters** of an assumed **probability distribution**, given some observed data.

Bayesian Regression

- MLE:

$$\arg \max_{\alpha, \beta, \sigma} \prod_{i=1}^n \mathcal{N}(y_i; \alpha + \beta x_i, \sigma)$$

- where $\mathcal{N}$ is a density function of normal distribution evaluated at y_i points, parametrized by means $\alpha + \beta x_i$ and standard deviation σ .
- In the Bayesian approach instead of maximizing the likelihood function, we would assume **prior distributions** for the parameters and use the Bayes theorem

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

$$\underbrace{f(\alpha, \beta, \sigma | Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i | \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_{\alpha}(\alpha) f_{\beta}(\beta) f_{\sigma}(\sigma)}_{\text{priors}}$$

Bayesian Regression

posterior $\propto$ likelihood $\times$ prior

$$\arg \max_{\alpha, \beta, \sigma} \prod_{i=1}^n \mathcal{N}(y_i; \alpha + \beta x_i, \sigma)$$

- The likelihood function is as above in MLE, however what changes in Bayesian approach is that we assume some prior distributions for the estimated parameters α, β, σ and include them into the equation

$$\underbrace{f(\alpha, \beta, \sigma | Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i | \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_{\alpha}(\alpha) f_{\beta}(\beta) f_{\sigma}(\sigma)}_{\text{priors}}$$

- While in MLE we look for a single optimal value for each of the parameters, in the Bayesian approach we obtain the **posterior distribution** of the parameters.

Mathematical Form (Bayes' Theorem)

- Where:
 - $P(\theta|\text{data})$ is the posterior distribution.
 - $P(\text{data}|\theta)$ is the likelihood function.
 - $P(\theta)$ is the prior distribution, which reflects your beliefs about θ before seeing the data.
 - $P(\text{data})$ is a normalizing constant.

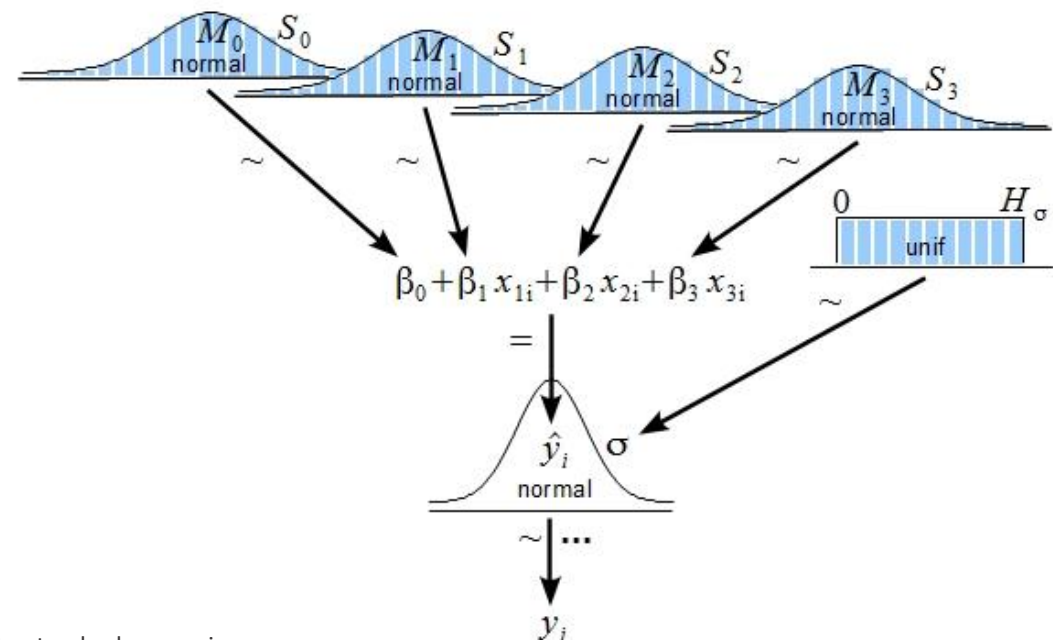
$$P(\theta \mid \text{data}) = \frac{P(\text{data} \mid \theta) \cdot P(\theta)}{P(\text{data})}$$

Bayesian Regression

- What distribution for priors?

- Unlimited number of choices.
- For example, assuming normal distribution for α, β
- For σ assume some prior distribution that is bounded to be greater than zero since σ needs to be positive.

$$\underbrace{f(\alpha, \beta, \sigma | Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i | \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_\alpha(\alpha) f_\beta(\beta) f_\sigma(\sigma)}_{\text{priors}}$$



Bayesian Regression

- The final estimate of posterior depends on the information that comes from your data and from your priors, but the more information is contained in your data, the less influential are priors.

$$\underbrace{f(\alpha, \beta, \sigma \mid Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i \mid \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_\alpha(\alpha) f_\beta(\beta) f_\sigma(\sigma)}_{\text{priors}}$$

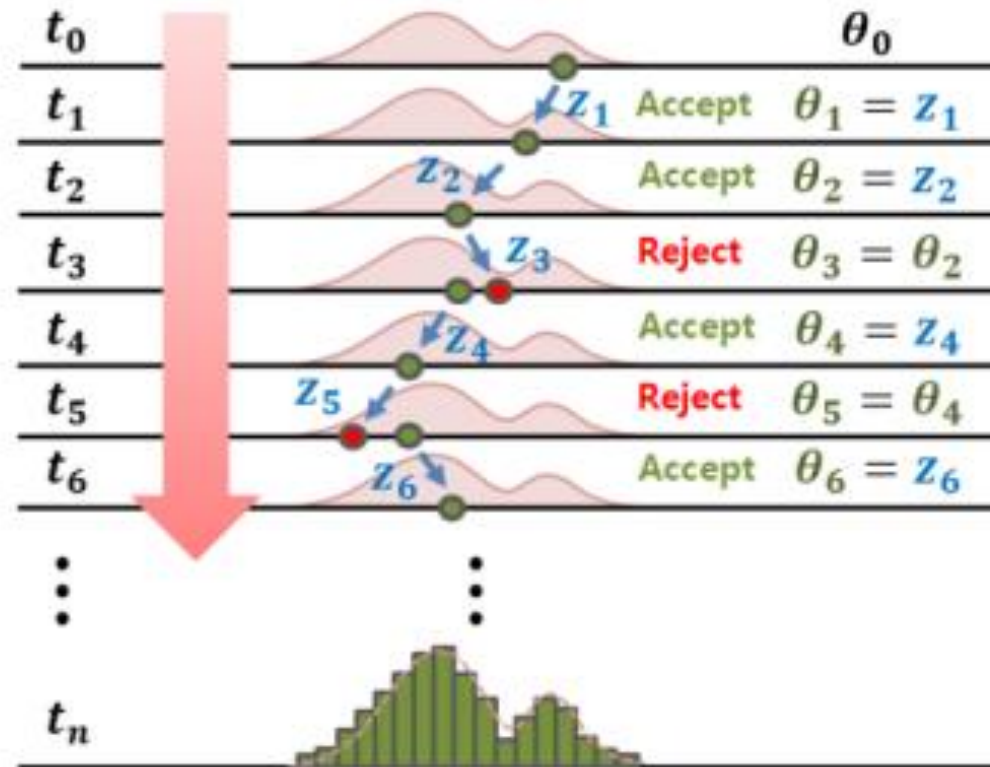
- If using uniform priors, since they are $f(\theta) \propto 1$, so the posterior distribution will reach its maximum at exactly the same point as the maximum likelihood estimate.
- In some cases, we can use **conjugate priors**, so the posterior distribution is directly available.
- However, in most cases, posterior distribution will not be directly available and we need to use Markov Chain Monte Carlo methods for estimating the model.

Conjugate Prior

- What is conjugate prior?
- “If the posterior distribution $p(\theta \mid \text{data})$ is in the same probability distribution family as the prior probability distribution $p(\theta)$, the prior and posterior are then called **conjugate distributions**, and the prior is called a **conjugate prior** for the likelihood function $p(\text{data} \mid \theta)$.”

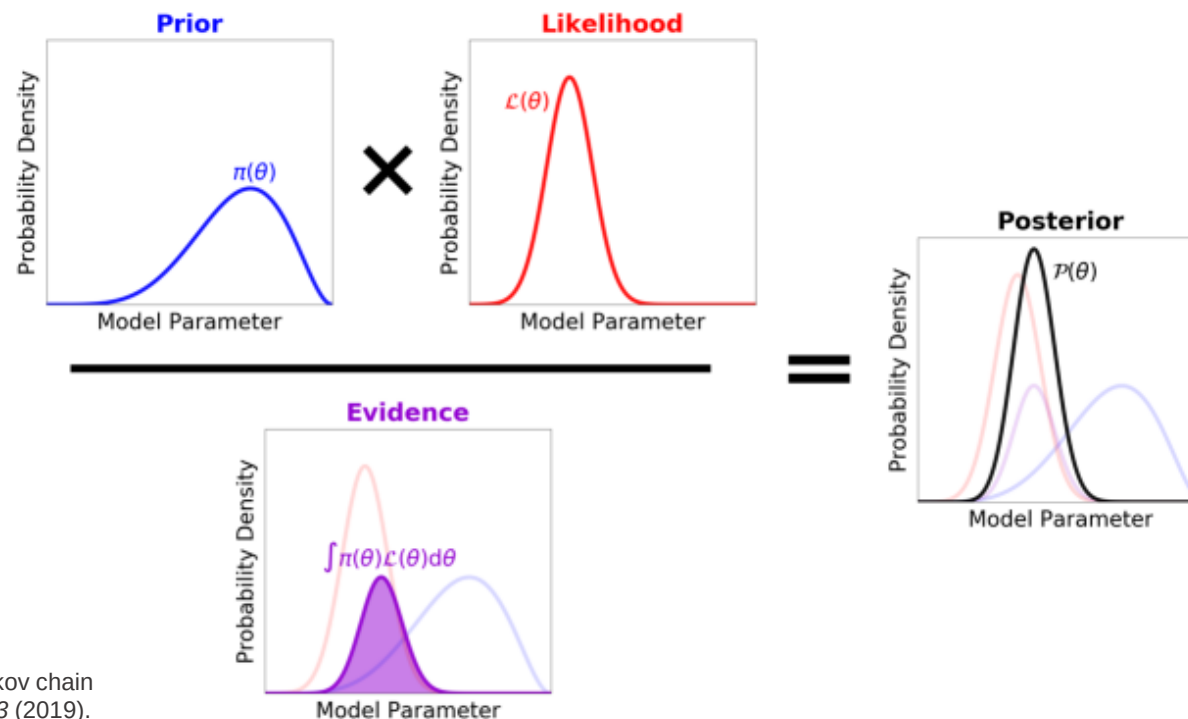
Illustration of Markov Chain Monte Carlo method

- MCMC: a class of algorithms for sampling from a probability distribution.



An illustration of Bayes' Theorem

- The posterior probability $P(\Theta)$ (black) of our model parameters Θ is based on a combination of our prior beliefs $\pi(\Theta)$ (blue) and the likelihood $L(\Theta)$ (red), normalized by the overall evidence $Z = \int \pi(\Theta)L(\Theta)d\Theta$.



Density Estimation

- Maximum likelihood (ML)
- Maximum a posteriori (MAP)
- Bayesian

The importance of probabilistic approach

- When learning some target function $f : X \rightarrow Y$, we can more generally learn the probabilistic function $P(Y|X)$.
- Using probabilistic approach allows us **designing algorithms that learn functions with uncertain outcomes** and incorporating prior knowledge to guide learning.
- Example:
 - Predicting tomorrow's stock price,
 - Prior knowledge: a bias that tomorrow's stock price is likely to be similar to today's price

Joint Probability Distributions

- **The key point in creating probabilistic models** is to define a set of random variables, and to consider the joint probability distribution over them.
- **Example:** a joint probability distribution over three random variables: a person's Gender, the number of HoursWorked each week, and their Wealth.

Gender	HoursWorked	Wealth	probability
female	< 40.5	poor	0.2531
female	< 40.5	rich	0.0246
female	$\geq$ 40.5	poor	0.0422
female	$\geq$ 40.5	rich	0.0116
male	< 40.5	poor	0.3313
male	< 40.5	rich	0.0972
male	$\geq$ 40.5	poor	0.1341
male	$\geq$ 40.5	rich	0.1059

The importance of knowing probability distribution

- Once we know the joint distribution we can answer every possible probabilistic question about these variables.
- if the joint probability distribution over an arbitrary set of random variables $\{X_1 \dots X_n\}$, is known then the conditional and joint probability distributions for arbitrary subsets of these variables (e.g., $P(X_n | X_1 \dots X_{n-1})$) can be calculated.
- This can be used to solve any classification, regression, or other function approximation problem defined over these variables, and furthermore produce probabilistic rather than deterministic predictions.
- **Example:** predict which people are rich or poor based on their gender and hours worked, $P(\text{Wealth} | \text{Gender}, \text{HoursWorked})$.

Learning the Joint Distribution

- How can we learn joint distributions from observed training data?
- Begin with a large database containing, say, descriptions of a million people?
- How the learning problem would change if we were to add additional variables to describe a total of 100 boolean features for each person?
- How many people on earth? approximately 10^{10} people

Learning the Joint Distribution

- To learn probabilities from available training data, we must be smart about
 - (1) how we estimate probability **parameters** from available data, and
 - (2) how we **represent joint probability** distributions.

Estimating Probabilities

- Example: flip a coin
- The coin is represented by the random variable X . If you flip this coin, it may turn up heads (indicated by $X = 1$) or tails ($X = 0$).
- The learning task is to estimate the probability that it will turn up heads; $P(X = 1) = ?$
- θ : the true (but unknown) probability of heads (e.g., $P(X = 1) = \theta$)
- $\hat{\theta}$: our learned estimate of the true θ .
- Let's gather training data by flipping the coin n times, and observe that it turns up heads α_1 times, and tails α_0 times.
- $n = \alpha_1 + \alpha_0$.

Estimating Probabilities

- What is the most intuitive approach to estimating $\theta = P(X = 1)$?

Probability estimation Algorithm 1: Given observed training data producing α_1 total "heads," and α_0 total "tails," output the estimate

$$\hat{\theta} = \frac{\alpha_1}{\alpha_1 + \alpha_0}$$